

Apple Sugar Contents Prediction via NIR-Style Image Generation: An Approach Based on an Extended CUT Model

Objective

This study aims to develop a deep learning model that predicts apple sugar contents by transforming RGB images into near-infrared (NIR)-style representations. This standard CUT (Contrastive Unpaired Translation) framework is extended with domain-specific loss functions that capture the physical characteristics of NIR imagery, such as reflectance and brightness distribution. The goal of this research is to validate whether the generated synthetic NIR images, when combined with an EfficientNet-based classifier, can provide accurate, non-destructive fruit quality assessment without the need for expensive NIR equipment.

Participants

[All participants' names]

¹Koh * Seo, ²Kim * Geon

1.Date: 2024 January

Participants: Koh Yun Seo, Kim Seung Geon

Topic: Development of AI Sugar Contents Prediction Sorting Device Using Citrus Peel Color

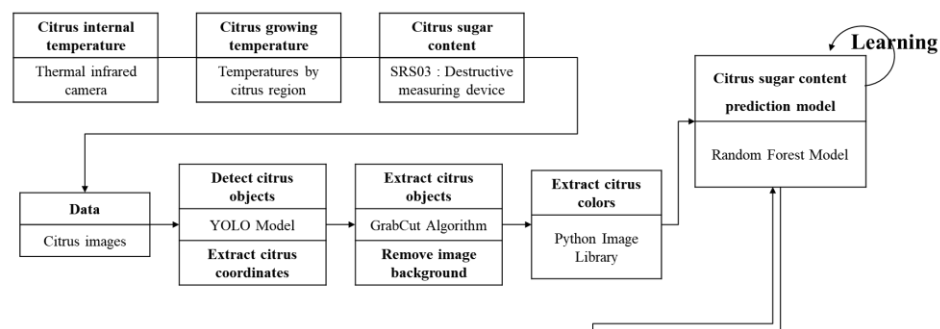
Objective:

- Estimate citrus sugar contents by AI using peel color.
- Analyze the RGB and HSV distribution of high-sugar citrus.
- Consider RGB and HSV values can be used to predict sugar contents if the citrus area can be segmented from an image and its color extracted.

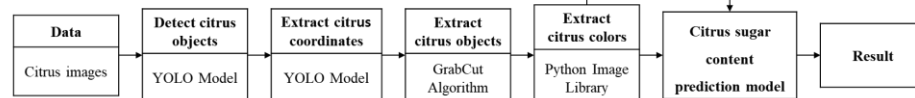
Contents:

1. Sugar Contents Prediction Model
 - A. **Citrus object detection [Citrus coordinate extraction]**
 - i. Using YOLO as an image-based object detection algorithm
 - ii. YOLO detects objects by predicting bounding boxes and class labels within the image
 - iii. Detects a citrus area in the image and extracts the coordinates of the citrus
 - B. **Extract citrus image [remove image background]**
 - i. Using the GrabCut algorithm, an image extraction algorithm for the foreground of an image
 - ii. Crop the square area containing the foreground
 - iii. Mark the background and missed foreground parts, and re-extract the image
 - C. **Extracting the color of a citrus**
 - i. Extract the color of a citrus using the Pillow(Python Imaging Library), a library for extracting colors from images in Python
 - D. **Implement a citrus sugar prediction model**
 - i. Using Random Forest Model, an ensemble learning method used for classification and regression in machine learning
 - ii. The Random Forest model can identify variable importance, maintains high accuracy even with missing values and outliers, and is less prone to overfitting

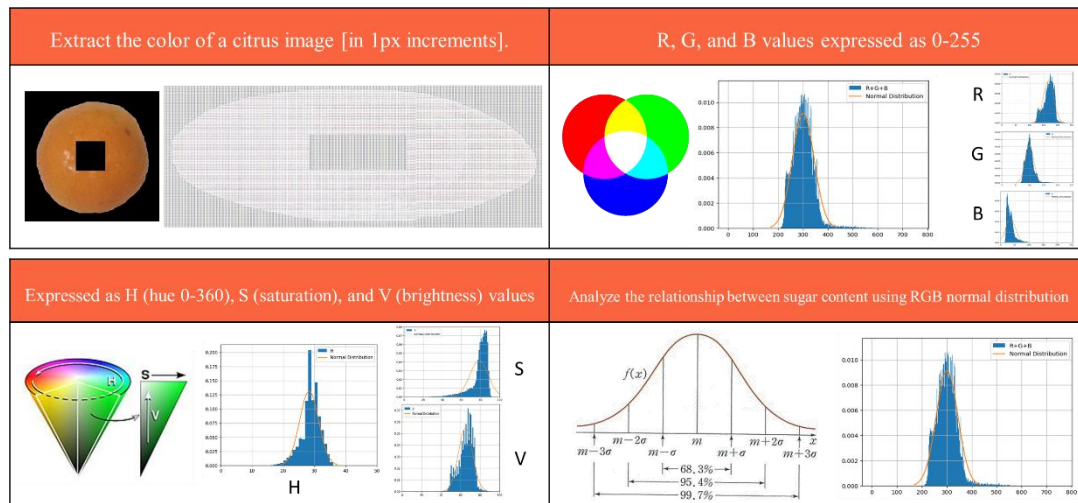
Stages of learning



Stages of prediction

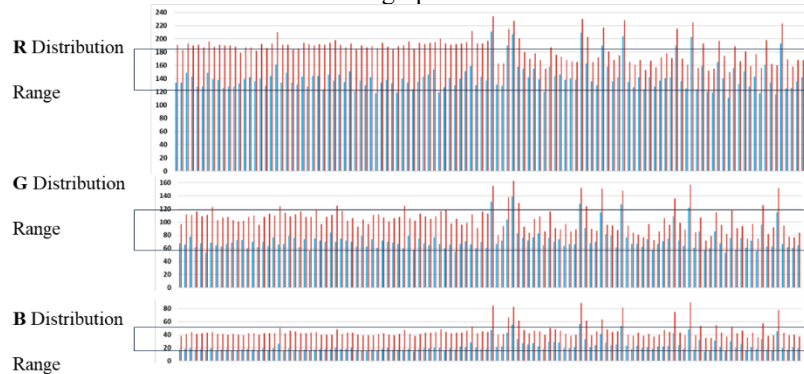


2. Extracting Color from a Citrus Image



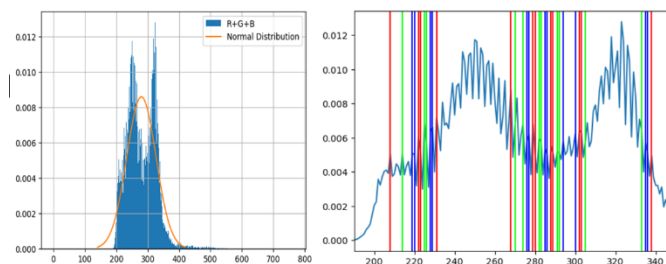
3. Analysis and Conclusions

A. 0~255 RGB value distribution graph



- R, G, and B have similar distribution ranges
- The results suggest the need of track the changes in color based on RGB values and analyze the relationship between these color variations and sugar contents.

B. Challenge: Non-normal Distribution in Combined RGB Values

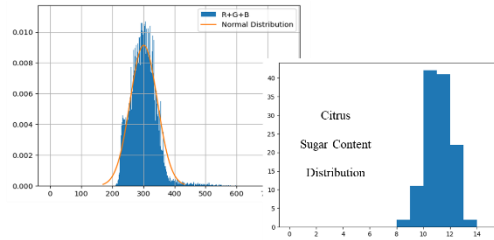


- Presence of value distributions that deviate from the normal curve
 - Select the central 80% of values with highest density
- C. Conclusions : Through experiments we realized that by analyzing correlation using certain type of citrus, we can predict sugar contents using a model trained on the color of citrus peels and sugar contents.

4. Sugar Contents Prediction Accuracy

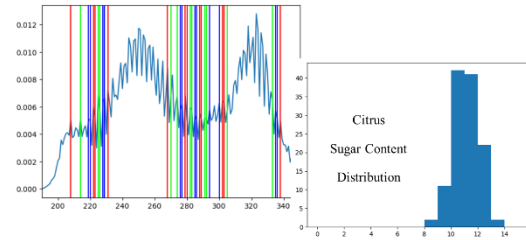
67%

- 120 data [100 training, 20 testing].
- Analyze the correlation between sugar content of the citrus and $-\sigma$ to σ values in the normal distribution of R+G+B



76%

- Extract the 80% range with a large distribution of R+G+B
- Analyze the relationship with sugar content



5. References

- Analyze the relationship between apple color and sugar contents using linear regression.
- Fruits and vegetables quality evaluation using computer vision

2.Date: 2024 February

Participants: Koh Youn Seo, Kim Seung Geon

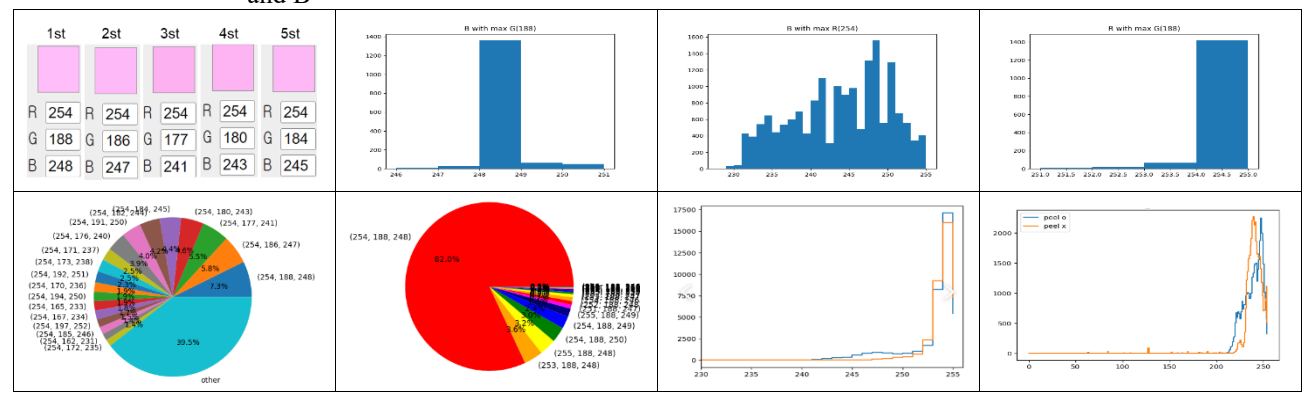
Topic: Problems Encountered During the Research and Addressing Them

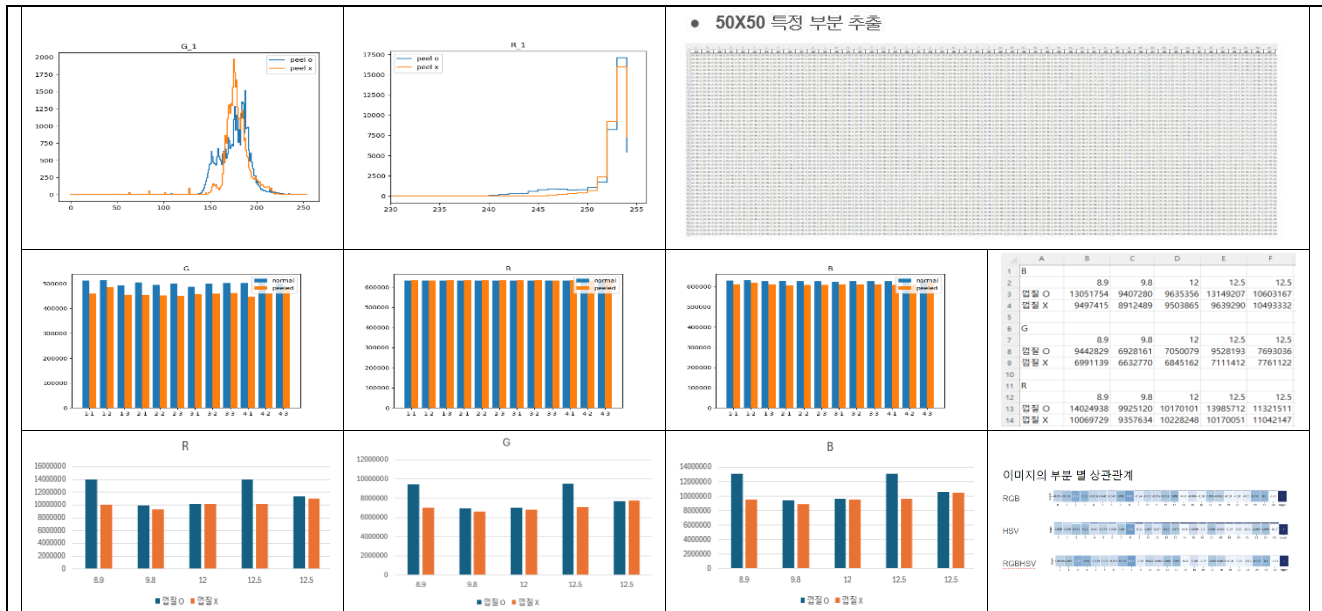
Objective:

- Analyze the relationship between the RGB and HSV values of citrus captured using an IR camera and their sugar contents
- Set the primary objective as identifying correlations between FT-IR spectroscopy results and IR camera data
- Investigate potential problems in tangerine image data captured by the IR camera

Contents:

1. Issues Related to the Experimental Environment
 - A. Issue with IR spectroscopy equipment: unable to access a spectrometer with its own light source
 - B. Conduct experiments using a portable visible–near-infrared (Vis–NIR) spectrometer for powdered material classification
 - C. Use an FT-IR spectrometer as a substitute for an NIR spectrometer
 - D. Determine that the FT-IR spectrometer is not fully suitable for this research
2. [Alternative] Experimental Environment Setup
 - A. Captured IR images of the tangerines used in the FT-IR spectroscopy experiments
 - i. Took images both with and without peels
 - B. Observed a high correlation in a different wavelength range compared to previous studies
 - i. Previous studies : 800 – 1800nm, Our study : 300 – 800nm(lower range)
 - ii. Determined the result was not significant
 - C. Extracted data within the same wavelength range (800–1800 nm) as in previous studies
 - i. Analyzed the result between the data and its sugar contents
 - ii. Developed the classification model based on the correlation
3. Experiment Results Analysis
 - A. Analyzed the relationship between the RGBHSV values of tangerines captured with an IR camera and their sugar contents
 - i. Performed data preprocessing and used Excel to conduct various forms of analysis, identifying correlations between RGB and HSV values and sugar contents
 - ii. Converted RGBHSV component values into color representations
 - iii. Represented data as partial sums of the overall RGBHSV values
 - iv. Analyzed the dominant color by identifying the component with the highest value among R, G and B





4. Hypothesis Based on Experimental Results

A. Problems in analyzing the correlation with sugar contents

- For the FT-IR spectrometer, Although an algorithm was used to express the correlation between the measured data and sugar contents, it was not clearly visible on the graph. There was also no formula to calculate sugar contents from the measured data.

- For the IR camera, it was also difficult to identify a clear correlation between the captured data and sugar contents.

C. Reformulation of the research hypothesis

- The hypothesis was revised to focus on identifying correlations between FT-IR spectroscopy results and IR camera image data.

- Analyzed potential issues in the tangerine data obtained from the IR camera

- Considered factors such as specialization of the measurement device, issues with background removal, data extraction and the accuracy of sugar contents measurement.

5. Reflection and Conclusion

This research began with a fundamental question: Is it possible to predict fruit sugar contents using only IR camera images? During the experimental preparation phase, we encountered difficulties stemming from the fundamental differences between FT-IR spectroscopy and IR imaging, as well as limitations in equipment availability. These challenges raised concerns about the reliability and repeatability of data captured by IR cameras. As a result, we formulated a new hypothesis: factors such as peel thickness, surface reflectivity, and internal fruit structure (e.g., fiber density, juice sac arrangement) may significantly affect the accuracy of sugar contents prediction based on IR images. To explore this hypothesis, we designed an experiment using tangerines under varying conditions, aiming to investigate how these physical characteristics influence the resulting image data and its relationship with measured sugar contents.

3.Date: 2024 March

Participants: Koh Youn Seo, Kim Seung Geon

Topic: Conduct the Experiment Again and Analysis Using 50 Tangerine Samples

Objective:

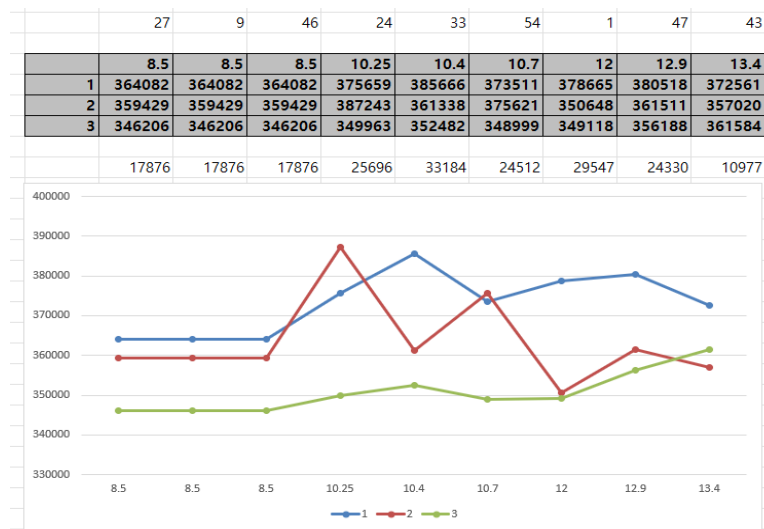
- Conduct experiment using 50 tangerine samples.
- Measure sugar contents three times and use the average of the second and third measurements
- Measure tangerines in three different forms: with peel, without peel, and peel only

Contents:

1. Experiment Using 50 Tangerine Samples
 - A. Captured 50 tangerine samples in three forms— with peel, without peel, and peel only—using an NIR camera
 - B. Measured sugar contents three times on the three different parts of each tangerine



- C. Although a clear correlation between sugar contents and the data was not observed, analysis of patterns revealed differences in IR reflectance depending on peel thickness



- D. Experiment result

- i. Additional factor to consider for to predict tangerine sugar contents: peel thickness

- The transmittance of the IR light is significantly affected by peel thickness in tangerines with varying peel thickness
 - Further research is needed on predicting tangerine sugar contents based on peel thickness
 - ii. Research on sugar contents prediction of fruits with thin peels is necessary
 - Using fruits with consistent peel thickness is expected to improve correlation analysis between FT-IR spectroscopy and IR camera images
 - iii. An absorber was installed on the inner walls of the darkroom shooting device to reduce the effects of IR light transmission, reflection, and refraction
 - E. Plan to shift the experiment to apple samples processed similarly to tangerines, focusing on fruit sugar contents prediction.
2. Reflection and Conclusion
- In this experiment, we captured IR images of 50 tangerines, both with and without peels, and measured their Brix values multiple times. The results revealed noticeable differences in reflectance across specific RGB and HSV channels; however, a strong numerical correlation with sugar contents levels was not observed. Notably, the presence or absence of the peel had a significant impact on reflectance patterns. These findings suggest that IR-based sugar contents prediction may be limited for fruits with thick peels and complex surfaces, highlighting the importance of physical characteristics such as peel thickness and surface reflectivity. Consequently, we began to shift our focus toward fruits with thinner peels and more uniform surfaces, such as apples, which may be more suitable for IR-based analysis. This direction was explored further in the following phase of our research.

4.Date: 2024.April
Participants: Koh Yun Seo, Kim Seung Geon
Topic: Research on a sugar contents prediction model using IR cameras.
Objective: ● Compare the correlation between image data captured by an IR camera and sugar contents values measured using an FT-IR spectrometer. ● Identify image features that show meaningful correlation with Brix values through various interpretations of IR images.
Contents: 1. Spectra obtained from tangerines using an FTIR spectrometer A. Experiment Procedure i. NIR images of tangerines were captured using an NIR camera and stored for analysis. ii. Brix values (sugar contents) were measured using a destructive sugar contents measurement device, and the results were recorded. iii. Analytical samples were prepared for spectral analysis of the tangerines iv. 10 satsuma mandarins and 10 Hallabong oranges are used and each tested in both peeled and unpeeled conditions. B. Spectral Analysis Result[선행 연구와 분석, 논문 제목 추가] i. Quantitative and qualitative significant data graphs from FT-IR spectral data : Key regions: 1700–1500 cm^{-1}, 1500–1300 cm^{-1}, 1100–950 cm^{-1} C. Interpretation of the Graph i. The correlation graphs from our tangerine experiments showed similar features to those in previous studies in the 1500–1300 cm^{-1} and 1100–950 cm^{-1} regions. ii. No clear features were observed or analyzed in the 1700–1500 cm^{-1} region. D. Based on these results, we conclude that the correlation between FT-IR spectra and sugar contents measured in this study can serve as a basis for characteristic analysis using IR cameras. E. Some less reliable results may be attributed to the limited number of tangerine samples used. Increasing the sample size is expected to yield more accurate outcomes.

2. Experiment on Tangerine Imaging Method Using NIR Camera

A. Experiment on Imaging Method Using IR Camera

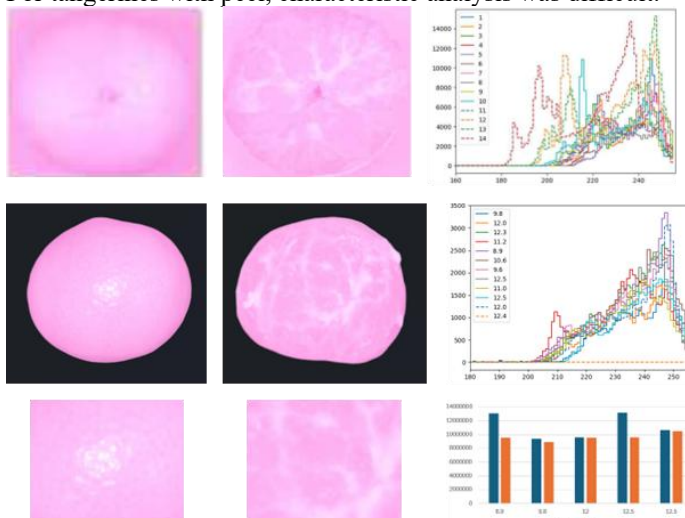
- i. Data analysis showed that meaningless values from the edges of the tangerines and areas without tangerines made it difficult to analyze characteristics.

B. RGBHSV Analysis After Cropping Tangerine Regions from the Images

- i. The extracted tangerine regions showed similar distribution patterns.
- ii. However, it was still difficult to find a correlation with sugar contents.
- iii. Differences in pixel size across images made value comparison challenging.

C. RGBHSV Analysis of Cropped Tangerine Regions in the Images

- i. For peeled tangerines, values tended to increase according to sugar contents.
- ii. For tangerines with peel, characteristic analysis was difficult.



- D. Using the 12 tangerine samples analyzed in section C, a sugar contents classification model was developed, achieving an accuracy of 0.66 and a loss of 1.09.

- E. Due to the limited number of samples, the results are somewhat constrained; increasing the sample size is expected to improve accuracy.

3. Theoretical foundation of fruit sugar contents measurement with infrared cameras

A. Correlation Analysis Elements

Solid sample spectra obtained using FT-IR spectroscopy (tangerine peel)

Liquid sample spectra obtained using FT-IR spectroscopy (tangerine juice)

Images of tangerines with peel taken by IR camera

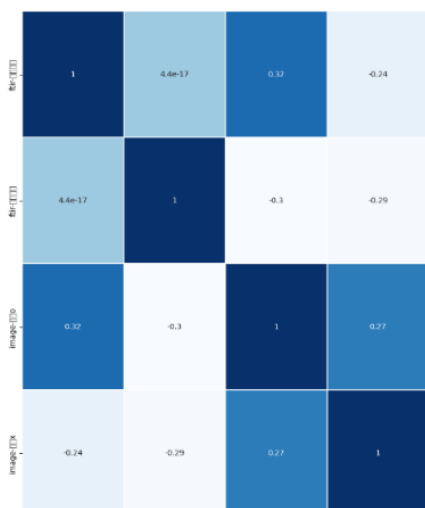
Images of peeled tangerines taken by IR camera

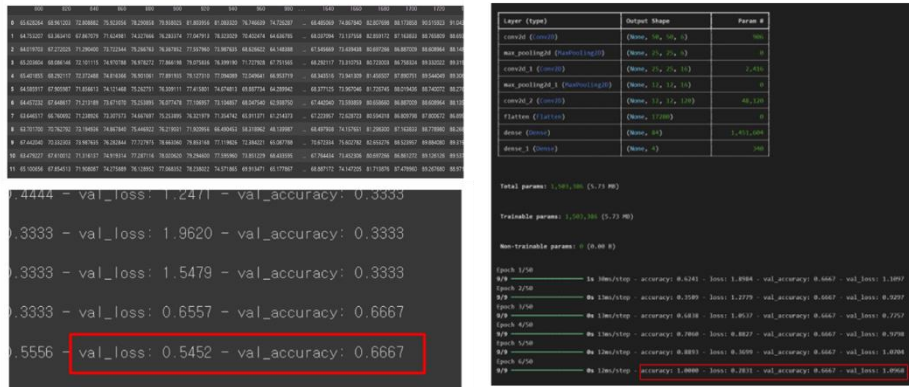
B. Correlation Analysis Result

- i. A high correlation was observed between sugar contents predicted by the classification model based on solid samples and sugar contents predicted using IR camera data of tangerines with peel.

- ii. A relatively high correlation was observed between sugar contents predicted by the classification model based on liquid samples and sugar contents predicted using IR camera data of peeled tangerines.

C. Based on these results, we conclude that predicting fruit sugar contents using IR camera data is feasible





FT-IR 분광 장치 측정 데이터를 이용한 분류 모델

- 정확도 0.66, 손실값 0.54

IR 카메라의 촬영 데이터를 이용한 분류 모델

- 정확도 0.66, 손실값 1.09

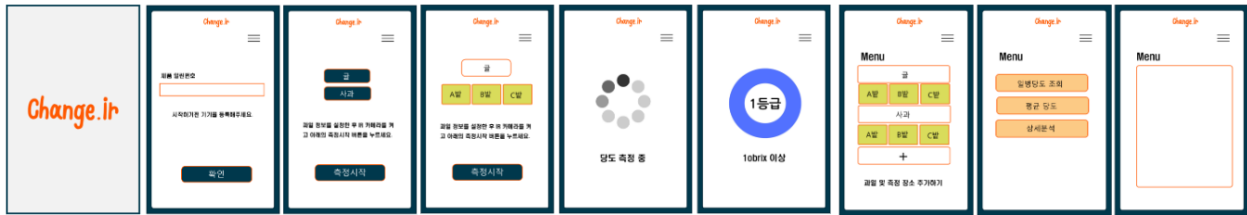
3. Reflection and Conclusion

- Based on the analysis of tangerine image data across RGBHSV channels, we observed variations in saturation and brightness values.
- However, the presence of noise and high variance from surface reflectance hindered a clear correlation with Brix values, limiting the reliability of IR camera-based sugar contents prediction. These findings highlighted the need to shift toward fruits with simpler structures and more consistent reflective properties to improve prediction performance.
- In particular, apples-with their thin peels and uniform surface reflectivity-were hypothesized to offer clearer patterns in NIR-style images for detecting sugar contents-related features.

5.Date: 2024 May
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Research on Fruit sugar contents Prediction Device and Model Using IR Camera
Objective: ● Construct on experimental space and setup for sugar contents measurement ● Develop the Change.ir learning model and app for fruit sugar contents prediction experiment. ● Capture images of 60 apples using an IR camera
Contents: 1. Experimental Space and Setup for Measurement A. NIR camera i. An NIR dimming device was installed ii. The NIR camera is capable of capturing both visible light and infrared images. B. Construct Experiment Space i. NIR camera is placed with external light blocked. ii. Absorptive materials and blackout curtains installed on the interior walls iii. NIR lighting adjusted and used for imaging in the absence of visible light C. sugar contents Prediction Experimental Device i. An automated imaging system was developed using a linear rail ii. A fruit sugar contents prediction device, modeled after a fruit sorting line, was developed to capture NIR images and predict sugar contents automatically D. Fruit object detection and sugar contents prediction model. i. Four AI models—LeNet-5, AlexNet, VGG, and Decision Tree—were implemented, trained on NIR and RGB images ii. A total of 22 features are available, and various combinations of models and features can be selected iii. The four models trained with selected features can be used for sugar contents prediction

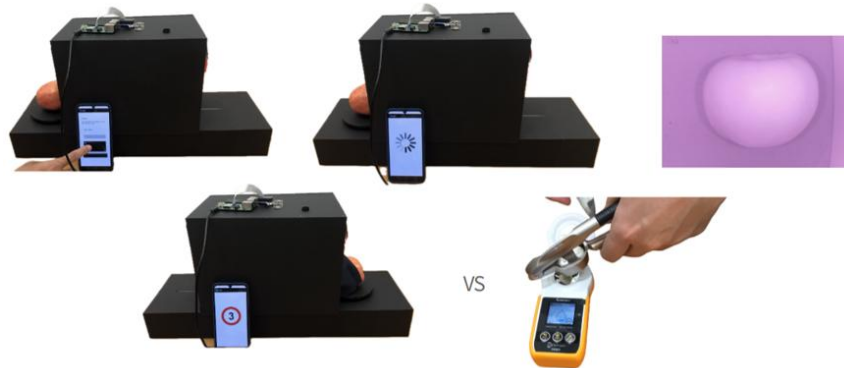
E. Development of the Change.ir app

- i. Allows users to set fruit information and display sugar contents measurement results.
- ii. Once measurement is complete, the results for the day are visualized in graph format
- iii. Users can view past fruit sugar contents data and check the average sugar contents



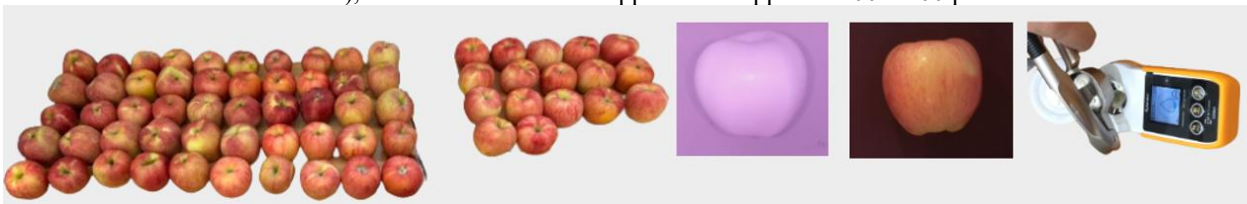
2. Procedure for Fruit Sugar Contents Prediction Experiment

- A. Set the fruit variety and measurement location before starting the test.
- B. When the “Start Measurement” button is pressed, the fruit enters the grading machine.
- C. Once the imaging is completed, the fruit’s grade is displayed on the app screen.
- D. The predicted sugar contents from our grading machine is compared with actual sugar contents values obtained from destructive testing to evaluate model performance.



3. NIR and Visible Light Imaging Experiment for Apple sugar contents Prediction Model.

- A. Captured images of 60 apples using the experimental setup with both NIR and visible light.
 - i. Acquired apple image data by capturing 60 apples with an NIR camera
 - ii. In the dark environment, NIR images were taken using the NIR lighting system.
 - iii. RGB images were taken under the same conditions with visible light turned on.
 - iv. All 120 images were preprocessed(object detection, background removal, coordinate and color extraction), and the center of each apple was cropped to 100 x 100 pixels.



B. Prediction results using NIR images

- i. Among the AI models in the Change.ir app, the Decision Tree model showed the best performance with approximately 80% accuracy.
- ii. Feature used : ['R', 'G', 'B', 'H', 'S', 'V', 'L', 'a', 'b', 'B_ir', 'H_ir', 'S_ir', 'V_ir', 'color_ir', 'size']

C. Conclusion

- i. Predicting fruit sugar contents through image analysis requires identifying consistent visual features.
- ii. These common features may vary depending on the fruit type, cultivation conditions, and variety.

4. Reflection and Conclusion

- i. This study aimed to establish a structured experimental environment for predicting fruit sugar contents by constructing an IR image acquisition system equipped with an automatic lighting control device.
- ii. To overcome the limitations identified in previous experiments using tangerines-such as inconsistent surface reflection and complex peel structures-we transitioned to apples as the primary subject of analysis.
- iii. Apples provide more uniform surface reflectance and a simpler structure, making them more suitable for image-based sugar contents prediction.
- iv. A total of 60 apples were imaged using both RGB and IR cameras under controlled lighting conditions, forming the foundation of a dataset for future deep learning experiments on NIR-style image translation and sugar contents prediction.

6.Date: 2024 June

Participants: Koh Youn Seo, Kim Seung Geon

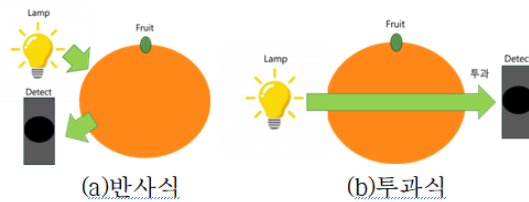
Topic: Background study on fruit quality evaluation models based on near-infrared (NIR) imaging.

Objective:

- Research is needed on converting RGB images to NIR images using image transformation techniques.
- Utilize NIR cameras equipped with NIR light sources.
- Based on captured NIR images, we aim to analyze the information about fruit quality and design a fruit quality evaluation model.

Contents:

1. Fruit Quality Evaluation Methodology
 - A. Principles of Near-Infrared Sugar Contents Measurement
 - i. Minhyuk So (2020), “Implementation of a Real-Time Non-Destructive sugar contents Measurement System and M-CNN Based Defect Detection Algorithm,” Cheongju University
 - ii. Non-destructive measurement using near – infrared light



[그림 1] 근적외선을 이용한 비파괴식 측정 방법]

- iii. Based on the positions of the light source and detector, the system can be classified into reflection and transmission types.
- iv. The reflection type illuminates the fruit surface and measures the reflected light.
- v. The system is designed to avoid direct reflection between light source and detector by adjusting the angle.
- vi. The reflection method enables measurement of a wide spectral range due to the short penetration path into the fruit surface.
- B. In our study, we aim to simulate a reflection-based approach by using NIR illumination projected onto the fruit surface and capturing the resulting NIR images using a NIR camera.
- C. Synthetic NIR Image Generation
 - i. Recently, RGB to NIR image translation techniques have been applied in various fields of computer vision.
 - ii. If synthetic NIR images can be generated from RGB images of fruits, they can be applied to fruit quality evaluation models. However, public fruit datasets are mostly limited to RGB images only.



deepNIR: Datasets for generating synthetic NIR images and improved fruit detection system using deep learning techniques Inkyu Sa, JongYoon Lim, Ho Seok Ahn and Bruce MacDonald

- D. Our research direction
 - i. Aim to construct a dataset containing RGB images, NIR images, and sugar contents data(Brix values) of fruits.
 - ii. Analyze the correlation between RGB and NIR images.
 - iii. Establish the foundation for generating synthetic NIR datasets to augment the dataset for model training.
2. Reflection and Conclusion
 - i. Due to the high cost and difficulty of collecting data using conventional NIR equipment, this study aims to explore the feasibility of generating NIR-style images from RGB inputs using GAN-based models.
 - ii. By utilizing these synthetic images, we seek to develop a fruit sugar contents prediction model that enables non-destructive quality assessment without the need for expensive sensors.

7.Date: 2024 July
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Feasibility Study on Predicting Fruit Sugar Contents through RGB-to-NIR Image Translation
Objective: ● Explore the feasibility of predicting fruit sugar contents using RGB and NIR images. ● Compare and evaluate this approach with conventional sugar contents prediction methods ● Consider the practical application and economic value of this research in the agricultural industry.
Contents: 1. Mechanisms and Feasibility of Technologies Used the Project. A. Conventional Methods for Predicting Sugar Contents(Non-Distructive Techniques) i. Measures the intensity of light reflected in the near – infrared spectrum to extract physical characteristics such as moisture, sugar contents and internal structure ii. Non-destructive sugar contents measurement is already used in actual farming practices. While accurate, these methods require expensive equipment and complex operational principles. B. NIR imaging i. NIR images are captured by detecting light reflected in the 700-1100 nm wavelength range. ii. Although they do not provide full spectral information, they can visualize the optical properties of materials. C. RGB to NIR image style translation i. RGB cameras cannot directly capture the NIR spectral range ii. However, indirect cues related to the NIR band can be inferred from light reflection and color patterns. iii. Deep learning models such as CycleGAN can learn these patterns and potentially convert RGB images into NIR style images. 2. Significance Compared to Conventional Sugar Prediction Methods. A. RGB to NIR translation is not a simple replacement for spectroscopy. Rather, it is an approach to indirectly infer spectral information using RGB image patterns. B. Research Significance i. By leveraging deep learning models to extract NIR related patterns from RGB images, this study presents a low cost, non-destructive alternative for predicting fruit sugar contents. 3. Industrial Applicability A. Advantages of Sugar Prediction Using RGB and NIR Images i. Offers improved efficiency and practicality compared to traditional methods. ii. Traditional tools like Brix meters and spectroscopic analyzers are precise but expensive, require skilled operators and often rely on destructive testing iii. Image-based prediction is non-destructive, allows for repeated measurements, and can be implemented with smartphones or simple sensors, making it highly economical. B. Easy Access to Large RGB Image Datasets i. https://www.aihub.or.kr/, onbuk Jangsu apple sugar contents quality data

- ii. Additional data collection is feasible at no extra cost using smartphones or agricultural drones.

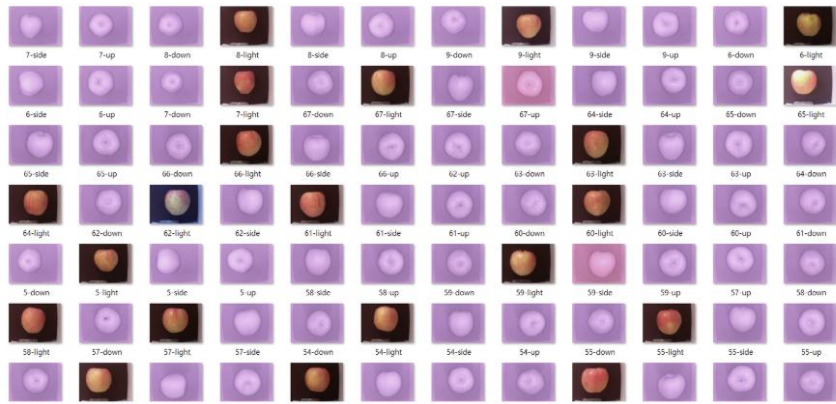
1. Training data - 2D RGB image data - Construction scale

Crop name	Non-destructive grade	Data count	ratio
Shinano Gold	A	17,207	3%
	B	20,581	4%
	C	5,586	1%
	subtotal	43,374	8%
Arisoo	A	11,651	2%
	B	22,692	4%
	C	7,787	1%
	subtotal	42,130	8%
Hongro	A	66,309	12%
	B	95,138	18%
	C	29,315	5%
	subtotal	190,762	36%
Fuji	A	127,509	24%
	B	87,734	16%
	C	44,182	8%
	subtotal	259,425	48%
total		535,691	100%

- C. Increasing viability of compact, low-cost NIR sensors for on- site deployment
- D. Enables real-time sugar contents prediction for farmers
- E. Supports diverse decision- making process, such as:
- Optimal harvest timing
 - Fruit grading and classification
 - Setting shipping and pricing standards
- F. Conclusion
- Potential for automating quality control throughout the production process.
 - Expected to yield direct economic benefits through labor savings and increased productivity.
 - Facilitates the selection of high quality produce and access to premium markets
 - Strengthens market competitiveness
4. Reflection and Conclusion
- he potential of using RGB images for fruit sugar contents prediction as an alternative to conventional NIR-based equipment, which often presents physical and cost-related limitations.
 - If NIR-style information can be inferred from RGB inputs using deep learning-based image translation models, it would enable the utilization of indirect spectral information without the need for expensive sensors.
 - This suggests that image-based prediction technology can be applied in agricultural fields in a non-destructive and cost-effective manner.
 - Furthermore, by leveraging RGB data easily acquired through general-purpose devices such as smartphones or drones, and combining it with NIR-style image generation models, this approach could support real-time quality prediction, shipment decision-making, and grade classification.

8.Date: 2024 August
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Project Plan for Designing and Developing a Sugar Contents Prediction Model Based on NIR Image Generation
Objective: ● Collect RGB images, NIR image data and sugar contents data. ● Select appropriate models for NIR image generation and sugar contents prediction ● Devise preprocessing techniques and feature extraction methods to enhance the predictive power of RGB and NIR image data
Contents: - Project Pipeline <pre> graph TD A[사과의 RGB 및 NIR 데이터 수집 및 전처리] --> B[이미지 생성 모델을 이용한 대량의 이미지 생성 및 성능 비교] B --> C[생성된 사과 모델을 이용한 분류 모델 학습 및 정확도 비교] C --> D[최적의 이미지 생성 및 당도 분류 방식 도출] D --> E[당도 예측 모델 구축] </pre> - Selection of Image Generation Model - CycleGAN(Cycle-Consistent Generative Adversarial Network) - Enables NIR image generation even when RGB and NIR images are not paired - A GAN based model that learns to translate between two domains using unpaired image data. - Utilizes two generators and two discriminators to translate images between domains. - Learns to preserve original image features through cycle consistency – translating back to the original domain should reproduce the source image. - Key Features of CycleGAN - Employs cycle consistency loss to ensure that transformed images remain similar to the originals. - Capable of training with unpaired datasets. - Applicable to various domains such as style transfer and restoration Limitation: Training can be unstable and time-consuming - CUT(Contrastive Unpaired Translation) - A GAN based model for unpaired image to image translation - Utilizes contrastive learning to achieve more precise transformations. - Unlike CycleGAN, CUT does not rely on cycle consistency loss. Instead, it preserves structural features across domains using contrastive loss - Key Features of CUT - Excels at maintaining unique patterns textures during image translation - Also capable of training with unpaired datasets - Offers more efficient and stable training compared to CycleGAN - Produces higher quality image translations and handles fine details effectively. - Data collection and Training Plan - Training Data for Image Generation Model

i. Directly capture RGB and NIR images of 60 apples



RGB, NIR images of 60 apples

- ii. RGB images: all images will be used for model training
 - iii. NIR images: Only side - view images that are advantageous for feature extraction will be used
- B. Training Data for Image Classification Model
- i. Utilize the trained NIR image generation model to convert a large RGB apple image dataset into NIR-style images
 - ii. The dataset, provided by the Jeonbuk Jangsu Apple sugar contents Quality Project (<https://www.aihub.or.kr>), was used for sugar contents prediction in this study
4. Reflection and Conclusion
- In this study, we aimed to generate NIR-style images using a large-scale RGB apple image dataset from AiHub. Since the dataset did not include paired NIR images, models suitable for unpaired image translation were required. Accordingly, CycleGAN and CUT-both widely used for unpaired image translation-were selected to examine whether they are suitable for generating NIR-style images from RGB inputs.

9.Date: 2024 September
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Theoretical review of the image generation model, CycleGAN
Objective: ● A mathematical understanding of the CycleGAN model ● Understanding of the Adversarial Loss ● Understanding of PatchNCE Loss
Contents: - Mathematical Understanding of CycleGAN - Final loss function $\mathcal{L}(G, F, D_A, D_B) = \mathcal{L}_{GAN}(G, D_B) + \mathcal{L}_{GAN}(F, D_A) + \lambda \mathcal{L}_{cyc}(G, F)$ - Adversarial Loss ($\mathcal{L}_{GAN}$) - The generator G(a) is trained to deceive the discriminator by producing images that appear realistic. - The discriminator is trained adversarially to distinguish real from fake images. - The loss encourages generators G and F, which map domains $A \rightarrow B$ and $B \rightarrow A$ respectively, to produce images that appear realistic in the target domain. - Cycle Consistency Loss ($\mathcal{L}_{cyc}$) - A loss that induces the input image to be similar to the original after translate between domains and then reconstruct - A loss that strengthens the preservation of original information during training. - Understanding of Adversarial Loss - Expectation $\mathbb{E}_{x \sim p(x)}[f(x)]$ calculates the expected value of the function f(x) over sampled drawn from the probability distribution p(x). - That is, it represents the average value of the function across the entire dataset. $\log D_B(b)$ - Discriminator (D_B) outputs the probability that the input image B belongs to the real domain B. - As a real number between 0 and 1, if $D_B(b) = 0.9$ the discriminator judges the image as real with 90% confidence. $\log D_B(b) = \log(0.9) = -0.105$ A lower loss value indicates better model performance. $\mathbb{E}_{b \sim p_{data}}(b)[\log D_B(b)]$ - The term that trains the model to classify real data as real - If $D_B(b) \rightarrow 1$ for a real B image b, it is recognized as real. $\mathbb{E}_{a \sim p_{data}}(a)[\log(1 - D_B(G(a)))]$ - The loss term that guides the model to distinguish generated images from real ones. - G(a) is a fake image generated by transforming an image from domain A into the style of domain B $D_B(G(a))$ should be close to 0 so that D_B classifies the generated image as fake. - Discriminator maximizes loss, Generator minimizes loss - Adversarial Loss - Generator G generates $A \rightarrow B$ images, Discriminator F discriminate real B and generated images. $\mathcal{L}_{GAN}(G, F) = \mathbb{E}_{b \sim p_{data}}(b)[\log D_B(b)] + \mathbb{E}_{a \sim p_{data}}(a)[\log(1 - D_B(G(a)))]$ - Understanding of Cycle Consistency Loss

- A. Prevention of domain A
 - i. $a \rightarrow G(a) \rightarrow F(G(a)) \approx a$
 - ii. When an image 'a' from domain A is transformed into domain B and then reconstructed back to domain A, the result $F(G(a))$ should be very similar to the original 'a'.
- B. Prevention of domain B
 - i. $b \rightarrow F(a) \rightarrow G(F(b)) \approx b$
 - ii. When an image 'b' from domain B is transformed into domain A and then reconstructed back to domain B, the result $G(F(b))$ should be very similar to the original 'b'.
- C. Cycle Consistency Loss
 - i. When $G(a)$ is reconstructed to $F(G(a))$, it should be close with original 'a', and opposite also should be same.
 - ii. Cycle Consistency Loss is the key component of CycleGAN
 - iii. Ensures meaningful image-to-image translation even under unsupervised learning conditions

$$\mathcal{L}_{cyc}(G, F) = \mathbb{E}_{a \sim p_{data}}(a) [\|F(G(a)) - a\|_1] + \mathbb{E}_{b \sim p_{data}}(b) [\|G(F(b)) - b\|_1]$$
- D. The reason why adversarial loss is important
 - i. CycleGAN doesn't require paired data.
 - ii. It is not sufficient for the translated image to merely appear realistic.
 - iii. A formula is needed to ensure meaningful bidirectional mapping.
4. Final loss function

$$\mathcal{L}(G, F, D_A, D_B) = \mathcal{L}_{GAN}(G, D_B) + \mathcal{L}_{GAN}(F, D_A) + \lambda \mathcal{L}_{cyc}(G, F)$$
 - A. Weight of loss function(λ)
 - i. 10 is commonly used to regulate the importance of original image reconstruction.
 - B. CycleGAN is for transforming unpaired data.
 - C. Two adversarial losses : $\mathcal{L}_{GAN}(G, D_B) + \mathcal{L}_{GAN}(F, D_A)$
 - D. Combine the cycle consistency loss $\lambda \mathcal{L}_{cyc}(G, F)$, which encourages the reconstruction of the original image.
 - E. Minimize $\mathcal{L}(G, F, D_A, D_B)$
5. Reflection and Conclusion
 - i. This study aimed to understand the mathematical structure and learning principles of the CycleGAN model prior to its practical application in experiments.
 - ii. CycleGAN is a representative model for unpaired image-to-image translation, with its core components-Adversarial Loss and Cycle Consistency Loss-working together to ensure both the quality and consistency of the image transformation.
 - iii. In particular, Cycle Consistency Loss helps minimize the loss of original information by reconstructing the RGB image after converting it into a NIR-style image, suggesting the potential for maintaining structural integrity and transferring style in future NIR transformations.

10.Date: 2024 October
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Implementation of a NIR image generation model using CycleGAN and a sugar content prediction model.
Objective: ● Implement an NIR image generation model using 60 RGB and NIR apple images. ● Generate fake images using the trained NIR image generation model. ● Sugar contents prediction model using NIR image data generated by CycleGAN
Contents: 1. Preparation for NIR image generating model construction A. Environmental setting i. Language / Framework: Python + PyTorch ii. Required package: pip install torch torchvision matplotlib iii. Operating Environment : Google Colab B. Dataset structure i. Usage of Unpaired Dataset ii. Folder structure - trainA/ (RGB image) - trainB/ (NIR image) - testA/ (RGB test image) - testB/ (NIR test image) iii. Resize each image to 256×256 C. Code implementation method i. Download the CycleGAN code repository - git clone https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix - cd pytorch-CycleGAN-and-pix2pix ii. Model training - Training command: python train.py --dataroot ./datasets/apple2nir --name apple2nir_cyclegan --model cycle_gan --gpu_ids 0 - Explanation of key parameters : --dataroot: specifying the data path, --name: setting the name, --model: specifying cycle_gan, --gpu_ids: GPU Usage status (0 = GPU) 2. CycleGAN training model creation and fake image generation model A. CycleGAN training model and fake image generation training model i. Train <pre>!python train.py \ --dataroot "/content/drive/MyDrive/datasets/apple2nir" \ --name apple2nir_cyclegan \ --model cycle_gan \ --gpu_ids 0 \ --n_epochs 25 \ --n_epochs_decay 25 \ --save_epoch_freq 5 \ --checkpoints_dir "/content/drive/MyDrive/checkpoints"</pre>

ii. Test

```
!python test.py \
--dataroot "/content/drive/MyDrive/datasets/apple2nir" \
--name apple2nir_cyclegan \
--model cycle_gan \
--phase test \
--checkpoints_dir "/content/drive/MyDrive/checkpoints" \
--gpu_ids 0
```

B. Visualize generated images

```
from PIL import Image
import matplotlib.pyplot as plt
fake_img = Image.open('/content/pytorch-CycleGAN-and-
pix2pix/results/apple2nir_cyclegan/test_latest/images/apple_fake_B.png')
plt.imshow(fake_img)
plt.title("Generated NIR-style Image (fake_B)")
plt.axis('off')
plt.show()
```



Images generated by CycleGAN trained on 60 apple RGB and NIR image

3. Implementation of a sugar contents prediction model using NIR image data generated by CycleGAN

A. Design of a regression model (CNN) for predicting fruit sugar contents

i. Flow Chart

- [1] RGB image
- [2] CycleGAN train
- [3] fake_B image generate
- [4] mapping Brix.csv(sugar contents data) and fake_B
- [5] Setup PyTorch Dataset
- [6] Train CNN regression model(fake_B -> predict sugar contents level)
- [7] Evaluation and visualization of prediction accuracy

ii. check root of fake images

```
/content/pytorch-CycleGAN-and-pix2pix/results/apple2nir_cyclegan/test_latest/images/
```


iii. Implementation PyTorch Dataset class

```

class BrixDataset(Dataset):
    def __init__(self, csv_file, image_dir, transform=None):
        self.data = pd.read_csv(csv_file)
        self.image_dir = image_dir
        self.transform = transform if transform else transforms.ToTensor()
    def __len__(self):
        return len(self.data)
    def __getitem__(self, idx):
        row = self.data.iloc[idx]
        base_filename = os.path.splitext(row['filename'])[0]
        fake_b_filename = f'{base_filename}_fake_B.png'
        img_path = os.path.join(self.image_dir, fake_b_filename)
        image = Image.open(img_path).convert('RGB')
        image = self.transform(image)
        brix = torch.tensor(row['brix'], dtype=torch.float32)
        return image, brix

```

iv. CNN regression model implementation

```

class BrixRegressor(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 16, 3, padding=1)
        self.conv2 = nn.Conv2d(16, 32, 3, padding=1)
        self.pool = nn.MaxPool2d(2, 2)
        self.fc1 = nn.Linear(32 * 64 * 64, 64)
        self.fc2 = nn.Linear(64, 1)
    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = x.view(-1, 32 * 64 * 64)
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x.squeeze()

```

B. Performance results of the CNN classification model

클래스	Precision	Recall	F1
Low	0.00	0.00	0.00
Mid	0.00	0.00	0.00
High	0.42	1.00	0.59

- i. Accuracy : 42%
- ii. The CNN model predicts all data as high

C. Prediction results of the CNN classification model

- i. The model predicted all samples as 'High', and it shows very poor classification performance.
- ii. This suggests that the sugar contents classification features are not reflected well in the fake_B images generated by CycleGAN.

D. Identification of the reason for the problem and directions for improvement

- i. The cause of the problem
 - The cause of the problem is that all images are transformed into the very similar style, and it prevents the CNN from detecting differences
 - The High class is the most frequent, so the model incurs relatively low loss by predicting only that class.
 - The model structure may be weak to extract high dimension features, or the amount of data can be insufficient.
- ii. Directions for improvement
 - By training a classification model using real NIR images instead of fake_B and building another model using CUT, we can compare their performance to identify the gap from the ground truth.
 - Adjusting the classification ranges can improve accuracy.
 - Implementing a different classification model instead of CNN may also enhance performance.

4. Conclusion

- i. In this study, we implemented a NIR-style image generation model using the CycleGAN framework, trained on 60 pairs of RGB and NIR apple images. The generated images(fake_B) were then used to develop a fruit sugar contents classification model.
- ii. CycleGAN, which enables image translation between unpaired domains, was considered suitable for generating NIR-style images from RGB inputs.
- iii. However, the CNN-based classification model predicted all samples as High, resulting in a low accuracy of 42%. This suggests that the fake_B images did not sufficiently reflect the features necessary for distinguishing sugar levels.
- iv. This issue is likely due to CycleGAN generating images with overly similar styles, making it difficult for the classifier to detect class differences.

Based on these results, future research will focus on comparing the performance with models trained on real NIR images to analyze the limitations of fake images. Furthermore, we plan to explore alternative image translation models such as CUT to improve classification performance.

11.Date: 2024 November
Participants: Koh Youn Seo, Kim Seung Geon
Topic: A Theoretical Review of the CUT Image Generation Model
Objective: ● A mathematical understanding of the CUT model ● Understanding of the Adversarial Loss ● Understanding of PatchNCE Loss
Contents: 1. A mathematical understanding of the CUT model A. Final loss function $\mathcal{L}(G, F, D_A, D_B) = \mathcal{L}_{GAN}(G, D_B) + \lambda \mathcal{L}_{PatchNCE}$ i. Adversarial Loss ($\mathcal{L}_{GAN}$) - The generator $G(a)$ is trained to deceive the discriminator by producing images that appear realistic. - The generator G conducts a one-way transformation from domain A to B. - Discriminator D_B is trained to discriminate real domain B image and $G(A)$. - The generator G is trained to make $G(A)$ appear like a real B image. - CUT has adversarial construction like CycleGAN, but it does not have F and D_A. ii. PatchNCE Loss ($\mathcal{L}_{PatchNCE}$) - PatchNCE Loss trains the model to keep the patch-level features of the converted image close to those of the original patch in feature space. - Induce natural transformation to the target domain style while preserving the semantic content of the input image, even without cycle consistency 2. Adversarial Loss(GAN Loss) : consistent with CycleGAN, but Adversarial Loss conduct one-way transformation A. $\mathbb{E}_{b \sim p_{data}}(b)[\log D_B(b)]$ i. Term that trains the model to classify real data as real ii. If $D_B(b) \rightarrow 1$ for a real B image b, it is recognized as real B. $\mathbb{E}_{a \sim p_{data}}(a)[\log (1 - D_B(G(a)))]$ i. Term that trains the model to classify fake data as fake ii. $G(a)$ is a fake image that transforms an image from domain A to the style of domain B. iii. Since D_B should recognize this image as fake, $D_B(G(a))$ should be close to 0 C. The discriminator is trained to maximize the loss, while the generator is trained to minimize it D. Adversarial Loss i. Generator G generates $A \rightarrow B$ image and Discriminator F discriminates real B generated images $\mathcal{L}_{GAN}(G, F) = \mathbb{E}_{b \sim p_{data}}(b)[\log D_B(b)] + \mathbb{E}_{a \sim p_{data}}(a)[\log (1 - D_B(G(a)))]/\tau$ 3. PatchNCE Loss $\mathcal{L}_{PatchNCE} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(f_q, f_k^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_q, f_k^j)/\tau)} \right]$ A. Each term of the equation i. f_q : query patch(feature of patch which is extracted from fake image) ii. f_k^+ : positive key(Corresponding patch feature at the same location in the real image a)

- iii. f_k^j : all keys (including negatives)
- iv. $\text{sim}(\cdot)$: cosine similarity
- v. τ : temperature (generally 0.07)
- vi. $\exp$: exponential function of softmax
- vii. $\mathbb{E}$: average of all patches

B. Reasons why we use cosine similarity

- i. The direction (structural meaning) of the feature vector is more important than its magnitude.
- ii. Similarity can be measured without information loss in the normalized feature space.
- iii. Direction information indicates shape, structure and pattern, and if shape and pattern (direction) are consistent, it is judged as similar.
- iv. CUT aims to change the style of an image while preserving its structure and content.

C. $\exp(\text{sim}(f_q, f_k^+)/\tau)$

- i. Score for transforming similarity between query and positive feature to softmax distribution
- ii. The higher the similarity, the larger the value of this term becomes, thereby increasing the likelihood of selecting positive pairs.

D. Softmax distribution

- i. A function that transforms real numbers to resemble probabilities

$$\text{softmax}(x_i) = \frac{\exp(x_i)}{\sum_j \exp(x_j)}$$

- ii. input x_i : score, similarity etc.
- iii. output : A probability between 0 and 1.
- iv. The reason for using Softmax in CUT
 - It can emphasize the highest value.
 - Softmax assigns higher probabilities to larger values and exponentially suppresses smaller ones.
 - It is suitable at emphasizing the most similar one among them.

E. The role of softmax in Contrastive Learning

- i. PatchNCE uses a softmax structure to learn to select positive pairs effectively.

$$\text{loss} = -\log \frac{\exp(\text{sim}(f_q, f_k^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_q, f_k^j)/\tau)}$$

- ii. Numerator : similarity of the positive pair
- iii. Denominator : sum of positive and negative pairs (normalization of the softmax distribution)
- iv. A training structure that maximizes the softmax probability of the positive pair focuses on the most similar candidate through softmax, and induce the model to the true positive.
- v. The model is trained to increase the similarity of positive pairs while decreasing the similarity with other candidates.

4. Final loss function

$$\mathcal{L}(G, F, D_A, D_B) = \mathcal{L}_{GAN}(G, D_B) + \lambda \mathcal{L}_{cyc}(G, F)$$

A. Weight of the loss term(λ)

- i. 10 is commonly used to balance the adversarial loss and the PatchNCE loss.

5. Reflection and Conclusion

- i. This study aimed to theoretically examine the learning principles of the CUT (Contrastive Unpaired Translation) model.
- ii. Unlike CycleGAN, CUT does not use Cycle Consistency Loss, but instead employs PatchNCE Loss to preserve the structural features of the input image while translating it into the target domain style.
- iii. The Adversarial Loss functions similarly to traditional GANs, while PatchNCE Loss ensures that each patch of the transformed image maintains feature similarity to the corresponding patch in the original image. This is achieved through contrastive learning, using cosine similarity and a softmax-based structure.

Through this theoretical analysis, it was confirmed that CUT has strong capabilities for structural preservation and enables natural style transfer. Based on these insights, future experiments will be designed using real image data to evaluate the performance of CUT in comparison to CycleGAN.

12.Date: 2024 December
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Performance comparison of sugar contents prediction models using CUT-based NIR image generation and CycleGAN
Objective: ● Generating Fake images using a NIR image generator model based on CUT ● Comparison of sugar contents prediction performance between CNN and CycleGAN models using CUT ● Comparison of sugar contents prediction performance using EfficientNet between CycleGAN and CUT models
Contents: 1. Preparation for NIR image generating model consturction A. Environmental setting i. Language / Framework: Python + Pytorch ii. Required package: pip install torch torchvision matplotlib iii. Operating Environment : Google Colab B. Dataset structure i. Usage of Unpaired Dataset ii. Folder structure - trainA/ (RGB image) - trainB/ (NIR image) - testA/ (RGB test image) - testB/ (NIR test image) iii. Resize each image to 256×256 2. Creation of CUT model and fake image generating model A. Creation of CUT model and fake image generating model i. Download CUT model <pre>git clone https://github.com/taesungp/contrastive-unpaired-translation.git cd contrastive-unpaired-translation</pre> ii. Train CUT model <pre>!python train.py \ --dataroot "/content/drive/MyDrive/datasets/apple2nir" \ --name cut_lambda_1.0 \ --CUT_mode CUT \ --gpu_ids 0 \ --n_epochs 20 \ --n_epochs_decay 20 \ --save_epoch_freq 5</pre> iii. Test CUT model <pre>!python test.py \ --dataroot "/content/drive/MyDrive/datasets/apple2nir" \ --name cut_lambda_1.0 \ --CUT_mode CUT \ --phase test \ --epoch latest \ --num_test 9999 \ --gpu_ids 0</pre>

B. Visualize the generated images

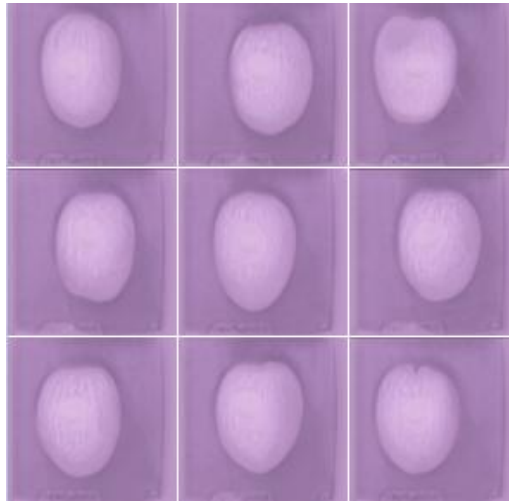


Image generated by CUT trained on 60 RGB and NIR images

3. Implementation of a sugar contents prediction model using NIR image data generated by CycleGAN

A. Design of a regression model (CNN) to predict fruit sugar contents

i. Flow chart

- [1] Collect RGB image
- [2] Train CUT → Generate fake_B image
- [3] Mapping fake_B image and Brix.csv
- [4] Construct PyTorch Dataset
- [5] Training CNN regression/classification model
- [6] Evaluate and visualize prediction accuracy

B. Performance results of CNN classification model

클래스	Precision	Recall	F1-Score
Low	0.00	0.00	0.00
Mid	0.00	0.00	0.00
High	0.42	1.00	0.59

- i. Accuracy : 42%
- ii. Predict all of Low, Mid, High as High

C. Prediction results of CNN classification model

- i. The model predicted all samples as 'High', resulting in very low classification performance
- ii. Generated fake_B image could not reflect difference of sugar contents, CNN model can not recognize it
- iii. A simple CNN model struggles to capture subtle style differences in the images

4. Performance comparison of CNN classification models using CycleGAN and CUT

모델	Accuracy	Low F1	Mid F1	High F1
CycleGAN	42%	0.00	0.00	0.59
CUT	42%	0.00	0.00	0.59

- A. Both models failed to reflect the actual Brix (sugar contents) distribution and predicted only 'HIGH'
- B. Fake_B images appear to lack meaningful information for Brix classification

5. Experiment Applying a Brix Prediction Model with Better Performance than CNN

- A. Our experiment aims to train a model with 60 apple images and analyze the performance of it.
- B. We plan to implement a classification model using EfficientNet, which can achieve accurate predictions with fewer parameters.
- C. EfficientNet reduces overfitting and improves performance on small datasets.
- D. EfficientNet can capture subtle differences in sugar contents in NIR-style images with its multi-expressiveness.

6. Performance comparison of EfficientNet classification models using CycleGAN and CUT

모델	Accuracy	Low F1	Mid F1	High F1
CycleGAN	40%	0.50	0.29	0.44
CUT	54%	0.40	0.33	0.67

- A. CycleGAN analysis
 - i. Low Class classification is relatively conducted well.
 - ii. Although EfficientNet has lower accuracy than CNN, it performs better in class balance.
- B. CUT analysis
 - i. EfficientNet showed a 12% improvement in accuracy compared to CNN.
 - ii. CNN cannot perform meaningful classification for both CycleGAN and CUT, showing only 42% accuracy.
 - iii. CNN failed in the Low and Mid classes and made predictions only in the High class.
 - iv. In the case of EfficientNet, predictions were excellent in all classes, showing 67% for the High class, 33% for the Mid class, and 40% for the Low class.
- C. CNN showed a collapse phenomenon, predicting all samples as belonging to the High class.
- D. EfficientNet showed dispersed predictions across classes

7. Conclusion

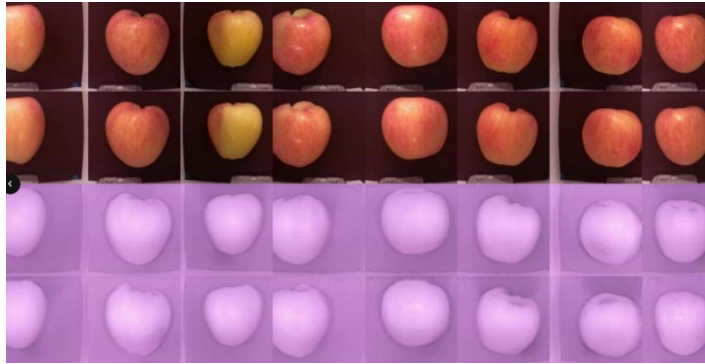
- i. We make a CUT-based image generation model and compared the performance of CNN and EfficientNet classifiers in predicting apple sugar contents using fake NIR images generated by both CycleGAN and CUT.
- ii. For the CNN classifier, both CycleGAN- and CUT-based fake images resulted in all samples being classified as High, with an accuracy of only 42%. This indicates that the generated fake images did not sufficiently reflect differences in sugar levels, and the simple CNN architecture was not capable of distinguishing subtle style variations.

Subsequently, we applied the EfficientNet model, which is known to perform well even with limited data. As a result, classification across all classes became possible. In particular, the CUT-based EfficientNet

model achieved 12% higher accuracy compared to the CNN model. F1-scores were 67% for the High class, 33% for Mid, and 40% for Low, confirming variability in class-wise prediction performance.

13.Date: 2025 January
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Comparative Study of Image Generation Algorithms for Improved Performance: CycleGAN, CUT, and Alternative Approaches
Objective: ● Explore the characteristics of the MUNIT model, which differs from CycleGAN and CUT. ● Build an image generation model using MUNIR and generate synthetic NIR images. ● Develop a sugar contents prediction model using EfficientNet based on the generated NIR images and to compare and analyze its performance
Contents: 1. Characteristics of CycleGAN and CUT A. Preservation of Apple Structure i. The models preserve the content of apple such as shape and size B. Domain Style Transformation i. NIR characteristics are incorporated through style transfer. C. Overall structure is maintained, resulting in high consistency across transformations. D. The purpose of converting RGB images to NIR images is to retain structure while altering fine textures and styles consistently 2. MUNIT Model (Multimodal Unsupervised Image-to-Image Translation (ECCV 2018)) A. MUNIT can generate multiple stylized outputs from a single input image, supporting one – to – many transformations B. Content and Style Disentanglement : The model separates an image’s representation into: i. Content : Structural elements such as apple shape and background layout ii. Style : Domain-specific elements such as lighting colortone, and NIR style iii. New styles can be generated through recombination. C. Bidirectional Translation i. Domain A(RGB) can be translated to domain B(NIR), and vice versa. ii. Enables the generation of diverse styles during the reverse transformation as well. D. Loss Functions i. Adversarial Loss : Basic GAN loss to encourage realistic outputs. ii. Reconstruction Loss : Maintains accuracy of content/style reconstruction iii. Style Loss : Maintains consistency in style distribution. 3. Building the Image Generation Model with MUNIT A. Model Building and Image Generation Process i. Experimental Setup: 60 RGB and 60 NIR images collected through direct photography. ii. Fake Image Generation: 30 RGB apple image used as input to generate synthetic NIR-style images.

iii. Final Output: Synthetic NIR images(fake images).



- B. Problems and Solutions
 - i. Style_encode() error: Resolved using encode()
 - ii. Single image processing limitation: Adjusted to process the entire testA folder.
 - iii. Networks.py recognition issue: Te-cloned from GitHub and verified manually.
4. EfficientNet Classification Model Based on Fake Images
 - A. Model Training
 - i. The Efficient Net model was trained to classify apple sugar contents(Brix) using the fake NIR images
 - B. Experimental Setup
 - i. Pretrained EfficientNet backbone from ImageNet
 - ii. Output classes : low, mid, high
 - iii. Input image size: 224*224
 - iv. Loss function: CrossEntropyLoss
 - v. Input images: 60 real RGB apple iages + 60 fake NIR images
 - vi. CSV mapping file: each apple's Brix value mapped to one of 3 classes
 - vii. Save path: /content/drive/MyDrive/MUNIT_outputs/apple2nir/images/
 - C. EfficientNet Model Configuration
 - i. Model
 - Created using create_model function from timm library
 - Backbone: EfficientNet_b0, pretrained on ImageNet
 - num_classes=3(Low, Mid, High)
 - ii. Criterion : nn.CrossEntropyLoss()
 - iii. Optimizer
 - torch.optim.AdamW (with weight decay
 - model.parameters()
 - lr=1e-4(learning rate)
5. EfficientNet Model Training Results
 - A. Test Performance Evaluation
 - i. Classification Report
 - Low : Precision: 0.00, Recall: 0.00, F1-score: 0.00
 - Mid : Precision: 0.27, Recall: 0.71, F1-score: 0.39
 - High : Precision: 0.38, Recall: 0.24, F1-score: 0.29
 - Overall Accuracy: 30%
 - B. Results Analysis
 - i. The training accuracy reached 100%, but test accuracy remained at 30%
 - ii. The majority of predictions were biased toward the Mid class
 - iii. The fake images may not have captured sufficient discriminative features required for accurate sugar contents classification.

6. Comparison with CycleGAN and CUT

모델	Accuracy	Low F1	Mid F1	High F1
CycleGAN	40%	0.50	0.29	0.44
CUT	54%	0.40	0.33	0.67
MUNIT	30%	0.00	0.39	0.29

A. Performance Comparison

- Compared to MUNIT, CycleGAN and CUT maintained better structural consistency while altering style elements

B. Conclusion

- It is advisable to implement the latest CUT-based models to achieve optimal performance for RGB – to- NIR conversion and subsequent sugar contents prediction

8. Reflection and Conclusion

- In previous studies, CycleGAN and CUT-based image generation models effectively preserved the overall structure of the image but failed to capture fine-grained features necessary for fruit sugar contents classification. As a result, the classification performance remained low. To address this limitation, this study explores the use of a structurally different image generation model—MUNIT—and evaluates its potential.
- MUNIT adopts a one-to-many architecture, allowing it to generate various styles of NIR images from a single RGB image. It separates image information into content (structural information) and style (domain-specific features), and recombines them to produce transformed outputs.
- Experimental results showed that the EfficientNet classifier trained on fake NIR images generated by MUNIT achieved 100% training accuracy, but only 30% test accuracy. Most test samples were predicted as the Mid class, and the Low class was not predicted at all, resulting in an F1-score of 0 for that class.

These findings indicate that MUNIT-generated images did not effectively express the features necessary for distinguishing sugar contents levels. Moreover, the model's ability to produce diverse styles may have hindered consistent classification performance.

14.Date: 2025 February
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Implementation of DCLGAN and DualCUT Models for Improved Performance in NIR Fake Image-based Brix Classification
Objective: ● Implement an EfficientNet classification model using NIR fake images generated by DCLGAN model and evaluate its performance. ● Implement an EfficientNet classification model using NIR fake images generated by the DualCUT model and evaluate its performance ● Compare the performance of the new models with previously studied EfficientNet classifiers based on CycleGAN, CUT, and MUNIT generated images
Contents: 1. Implementation of EfficientNet Classifier Using NIR Fake Images Generated by DCLGAN A. DCLGAN (Dual Contrastive Learning GAN) i. An improved version of CUT that applies two levels of contrastive learning simultaneously to enhance performance ii. Local contrastive loss : Similar to CUT, applies patch-level feature comparisons. iii. Global contrastive loss : Learns feature similarity at the level of entire global image representations. iv. Enables more precise style transfer in cases requiring complex and combined style transformation. B. Environment Setup i. Language/Framework: Python + PyTorch ii. Required packages: torch, torchvision, matplotlib, pandas C. Dataset Configuration i. Unpaired Dataset used ii. Image size: Resized to 256x256 D. Code Implementation i. Official DCLGAN repository: git clone https://github.com/ssu-hpc/DCLGAN ii. Dependency installation: pip install -r requirements.txt E. DCLGAN training and fake image generation i. Training command: <pre>!python train.py \ --dataroot "/content/drive/MyDrive/datasets/apple2nir" \ --name apple2nir_dclgan \ --model dclgan \ --CUT_mode DCL \ --gpu_ids 0 \ --n_epochs 25 \ --n_epochs_decay 25 \ --save_epoch_freq 5 \ --checkpoints_dir "/content/drive/MyDrive/checkpoints"</pre> ii. Testing(fake image generation): F. EfficientNet Classification Using Fake NIR Images i. Brix.csv mapped with corresponding fake NIR images. ii. Dataset configured for EfficientNet input. iii. EfficientNet classification model trained and evaluated. G. Performance Evaluation of EfficientNet Classifier

Class	Precision	Recall	F1-score
Low	0.57	0.67	0.62
Mid	0.00	0.00	0.00
High	0.33	0.14	0.20

- i. Accuracy : 38%
 - ii. Mid class is not detected
 - iii. Only 1 out of 7 High class samples correctly predicted.
 - iv. Although image structure was preserved, the generated spectrum-based features were insufficient for effective classification.
2. Implementation of EfficientNet Classifier Using NIR Fake Images Generated by DualCUT
- A. DualCUT (Contrastive Unpaired Translation with Attention)
 - i. Extended CUT model with an attention module added.
 - ii. Incorporates CBAM (Convolutional Block Attention Module) or SE (Squeeze and Excitation) blocks within the Generator to enhance focus on important features.
 - B. Code Implementation
 - i. Based on CUT repository: `git clone https://github.com/taesungp/contrastive-unpaired-translation`
 - ii. Attention module added to `networks.py` or implemented in `modules/attention.py`.
 - iii. Typical insertion points: within the Generator's ResBlock or Encoder layers.
 - C. CUT – based Model Implementation
 - i. ChannelAttention, SpatialAttention and CBAM modules implemented as follows:


```
class ChannelAttention(nn.Module):
    def __init__(self, in_planes, ratio=8):
        super(ChannelAttention, self).__init__()
        self.avg_pool = nn.AdaptiveAvgPool2d(1)
        self.max_pool = nn.AdaptiveMaxPool2d(1)
        self.fc = nn.Sequential(
            nn.Conv2d(in_planes, in_planes // ratio, 1, bias=False),
            nn.ReLU(),
            nn.Conv2d(in_planes // ratio, in_planes, 1, bias=False)
        )
        self.sigmoid = nn.Sigmoid()
    def forward(self, x):
        avg_out = self.fc(self.avg_pool(x))
        max_out = self.fc(self.max_pool(x))
        return self.sigmoid(avg_out + max_out)

class SpatialAttention(nn.Module):
    def __init__(self):
        super(SpatialAttention, self).__init__()
        self.conv = nn.Conv2d(2, 1, kernel_size=7, padding=3, bias=False)
        self.sigmoid = nn.Sigmoid()
    def forward(self, x):
        avg_out = torch.mean(x, dim=1, keepdim=True)
        max_out, _ = torch.max(x, dim=1, keepdim=True)
        x_cat = torch.cat([avg_out, max_out], dim=1)
        return self.sigmoid(self.conv(x_cat))

class CBAM(nn.Module):
    def __init__(self, in_planes):
        super(CBAM, self).__init__()
        self.ca = ChannelAttention(in_planes)
        self.sa = SpatialAttention()
```

```
def forward(self, x):
    x = self.ca(x) * x
    x = self.sa(x) * x
    return x
```

D. DualCUT Training and Fake Image Generation

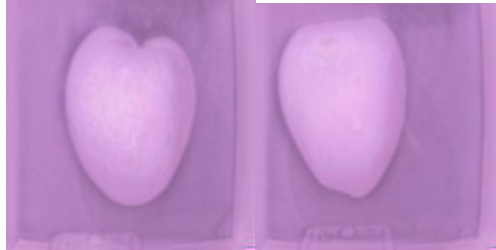
i. Training command:

```
!python train.py \
--dataroot /content/drive/MyDrive/datasets/apple2nir_rgb \
--name apple2nir_dualcut \
--model cut \
--dataset_mode unaligned \
--input_nc 3 --output_nc 3 \
--direction AtoB \
--nce_idt \
--lambda_NCE 1.0 \
--nce_layers '0,4,8,12,16' \
--netG resnet_9blocks \
--netF mlp_sample \
--load_size 256 --crop_size 256 \
--gpu_ids 0 \
--batch_size 4 \
--lr 0.0002 \
--n_epochs 100 --n_epochs_decay 100
```

ii. Testing(fake image generate)

```
!python test.py \
--dataroot /content/drive/MyDrive/datasets/apple2nir_rgb \
--name apple2nir_dualcut \
--model cut \
--phase test \
--no_dropout
```

E. EfficientNet Classification Using Fake NIR Images



F. Performance Evaluation of EfficientNet Classifier

Class	Precision	Recall	F1-score
Low	0.38	1.00	0.55
Mid	0.00	0.00	0.00
High	0.00	0.00	0.00

i. Accuracy : 38%

ii. The model classified almost all images as Low class.

iii. High and Mid classes were no detected at all(class imbalance issue).

3. Performance Comparison with Precious Models

Model	Accuracy	Low F1	Mid F1	High F1
Real NIR	50%	0.00	0.33	0.50
CycleGAN	40%	0.50	0.29	0.44
CUT	54%	0.40	0.33	0.67
MUNIT	30%	0.00	0.39	0.29

	DCLGAN	38%	0.57	0.00	0.17
	DualCUT	38%	0.55	0.00	0.00

A. Analysis of DCLGAN Results

- Although DCLGAN leverages both local and global contrastive loss, it was less effective in fine-grained class discrimination
- The global contrastive loss may have interfered with the learning of subtle features.
- Mid – class features were not sufficiently captured, suggesting the need for additional feature disentanglement of attention mechanisms.

B. Analysis of DualCUT Results

- The combination of attention modules with PatchNCE and DualCUT may have led to over-attention effects.
- The model architecture became more complex, and performance suffered from imbalance
- The attention mechanism possibly caused overfitting to foreground features, resulting in biased learning toward the Low class and complete neglect of Mid and High classes.

4. References

- DCLGAN: Dual Contrastive Learning for Unsupervised Image-to-Image Translation, CVPRW 2021
- Multi-attention bidirectional contrastive learning method for unpaired image-to-image translation, PLOS ONE, 2024
- EfficientNet: Tan et al., "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks", ICML 2019
- <https://github.com/qlc2018/DCLGAN-PyTorch>
- <https://github.com/rwightman/pytorch-image-models> (timm)

5. Reflection and Conclusion

- In previous experiments using various image translation models (CycleGAN, CUT, and MUNIT), the CUT model demonstrated the most promising classification performance. Based on this finding, the present study investigates the potential for further improvement by testing two recently proposed CUT-based models: DCLGAN and DualCUT.
- DCLGAN incorporates both local and global contrastive losses to learn the overall structure of images in greater detail.
- However, experimental results showed that its classification accuracy remained at 38%, with the Mid class not being predicted at all. This suggests that the use of global contrastive learning may have interfered with the learning of fine-grained features, making it difficult to disentangle the complex characteristics of the Mid class.
- DualCUT, on the other hand, combines the CUT architecture with an attention module to emphasize important regions during training. Nonetheless, most samples were classified as belonging to the Low class, resulting in an overall accuracy of 38%, while the Mid and High classes were not predicted at all. This outcome indicates that the attention mechanism may have overly focused on features associated with a single class, thereby disrupting class balance in classification.

Although both models are structurally enhanced compared to CUT, their actual classification performance was lower than or comparable to the original CUT model. A major limitation observed in both models was the inability to effectively represent Mid and High classes, leading to class imbalance in predictions.

15.Date: 2025 March
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Second Implementation of the DualCUT Model and Performance Evaluation of EfficientNet Classifier Using Re-generated NIR Fake Images.
Objective: ● Analyze and reflect on why the DualCUT model performed worse than the baseline CUT model ● Identify potential omissions or issues in the initial implementation and conduct a second round of experiments ● Re-evaluate and compare the classification performance with previously studied image-based EfficientNet models
Contents: 1. Comparative Analysis of DualCUT and CUT A. Theoretical Background of Performance Differences i. CUT : Based on contrastive learning using PatchNCE loss ii. DualCUT : Extends CUT by incorporating CBAM(Spatial Attention and Channel Attention) into the architecture. iii. Based on theory, DualCUT should enable more precise image translation by focusing on important regions of the image. B. Potential Reasons for Lower Performance of DualCUT i. Adding attention modules can inhibit gradient flow or cause model to focus on only a narrow portion of the image, reducing generalization. ii. If the attention mechanism over-emphasizes a small subset of patches, it may degrad overall performance. iii. CBAM reflects visual saliency, which may be suboptimal for fruit images with subtle structural variations iv. CBAM is a conditional enhancement; when not properly tuned of inserted at the appropriate layer, it can distort feature representations and degrade performance. C. Potential Advantages of DualCUT in Fruit Classification and NIR Translation. i. Model Structure: - CUT with added Attention modules and bidirectional contrastive learning. ii. Spatial Attention: Learns which regions are important based on channel-wise average and max pooling of feature maps. - Often enhance edges, contours, and regions of high contrast iii. Bidirectional PatchNCE Loss: Encourages learning of important regions in both source and target domains. iv. Findings from other DualCUT experiments: - Enhanced edge sharpness – clearer separation of foreground and background - Improved preservation of object contours(e.g., people, apples, animals) - Background appears smoother while the foreground remains sharp after translation - CBAM suppresses irrelevant features, leading to a more targeted translation. v. Advantages for NIR Domain: - NIR images emphasize material reflectance rather than color. - Apples exhibit different reflectance compared to the background and vary across the surface due to curvature and texture. - Clear boundaries between object and background facilitate better spectral domain translation. vi. Emphasizing apple foreground structure is crucial for meaningful NIR translation - DualCUT can preserve global apple structure, reflectance patterns, and surface texture more faithfully during translation. D. Based on these insights, we conducted a second DualCUT implementation and experiments 2. Second Implementation of DualCUT and Experiments

A. Verification of Potential Omissions and Second Implementation

- i. Architecture Summary: CUT with CBAM enhanced ResNet blocks.
- ii. Steps:
 - Download CUT repository
 - Add CBAM Attention module
 - Replace Generator Resnet blocks with ResnetBlockWithCBAM
 - Train the model and generate fake NIR images



B. EfficientNet Classification Performance Evaluation

i. First classification Model Results

Class	Precision	Recall	F1-score
Low	0.00	0.00	0.00
Mid	0.60	0.60	0.60
High	0.60	0.75	0.67

- Accuracy : 60%
- All Low-class samples misclassified and Mid
- Some boundary confusion between Mid and High classes observed.

ii. Second Classification Model Results

Class	Precision	Recall	F1-score
Low	0.00	0.00	0.00
Mid	0.43	0.60	0.50
High	1.00	0.25	0.40

- Accuracy : 40%
- Low class again misclassified as Mid
- High class predictions were few but highly accurate.

iii. Third Classification Model Results

Class	Precision	Recall	F1-score
Low	0.00	0.00	0.00
Mid	0.62	1.00	0.77
High	1.00	0.50	0.67

- Accuracy : 70%
- Perfect recall and high F1-score for Mid class
- High class had high precision, but some misclassified as Mid
- Low class remained unrecognized

3. Comparative Analysis with Previous Image-based EfficientNet Models

A. Observations on DualCUT vs CUT

: Embedding attention mechanisms in a contrastive learning-based model like CUT may cause the model to overemphasize specific features, leading to good training performance but poor generalization on new data

B. Impact of Domain Gap

: When working with fake NIR images, where the domain gap is large and the data is sensitive, excessive attention can distort critical information instead of improving it.

C. Model Complexity

- : The generator architecture becomes over-complicated with additional attention blocks, potentially destabilizing training or impairing gradient flow
- D. Sensitivity to Feature Extraction Layer Selection
 - : The placement of PatchNCE feature extraction layers significantly impacts performance – improper selection can lead to suboptimal results
- E. Sensitivity to Feature Extraction Layer Selection

4. Reflection and Conclusion

- i. To investigate the causes behind the lower-than-expected performance of the original DualCUT model, this study reimplemented the DualCUT architecture and conducted experiments to evaluate its potential for more effective learning. In particular, both theoretical analysis and empirical evaluation were conducted to assess the impact of the CBAM attention module on model performance.
- ii. DualCUT is essentially a modified version of the CUT architecture, with the CBAM module integrated to enhance the precision of style translation by emphasizing important spatial and channel-wise information. Theoretically, this approach is expected to improve the quality of NIR-style image generation, especially for objects like apples where reflective boundaries are crucial, by strengthening foreground-background separation.
- iii. However, initial implementation experiments revealed that the attention module caused the model to over-focus on specific patches, leading to poor generalization across the entire image. In all three classification experiments using EfficientNet, the Low class was never predicted, and while the Mid class showed reasonable results, the High class exhibited inconsistent prediction accuracy.
 - 1st Experiment: Accuracy 60%, High class F1-score 0.67
 - 2nd Experiment: Accuracy 40%, correct but sparse predictions for High class
 - 3rd Experiment: Accuracy 70%, perfect prediction for Mid class, partial misclassification for High class

These results suggest that the attention mechanism may have overemphasized certain regions, thereby hindering balanced feature learning across the entire image. This issue is particularly problematic for fake NIR images, which inherently involve domain gaps and sensitive spectral characteristics. If attention mechanisms are not carefully regulated in such contexts, they may cause feature distortion, ultimately leading to a decline in classification performance.

16.Date: 2025 April
Participants: Koh Youn Seo, Kim Seung Geon
Topic: Implementation of a new image generation model by modifying the structure of the CUT model
Objective: ● Structural modification of the CUT model from mathematical aspect ● Implementation of a new image generation model based on CUT and generated fake images ● Performance analysis and evaluation using the EfficientNET classification model
Contents: - Modified the Structure of the CUT Model from the Perspective of Mathematically Extending CUT - Add Domain-Specific Loss (physics-based correction loss) to the existing CUT - Add an additional loss term which reflects feature of NIR domain, maintaining CUT structure. - Reflectance Smoothness Loss - NIR is based on wavelength reflectance: internal tissue patterns are often more important than boundaries. - Introduce a local uniformity penalty. - Histogram Matching Loss - Histogram : the distribution of pixel values within the image - Add a histogram-matching loss to ensure a brightness distribution similar to that of NIR - This can complement problem that it tends to ignore tone and contrast, maintaining structure of CUT. - Relevance of the Domain-Specific Loss to the characteristics of NIR - Histogram Matching Loss : NIR can reflect the distinct average brightness. - Reflectance Smoothness Loss : In NIR images, representing internal material structures is crucial, so it is possible to maintain internal tissue consistency based on reflectance characteristics. - DCT/Frequency Domain Loss : reflectance patterns can also be analyzed in the frequency domain. - A study incorporating physics-based correction loss into the CUT framework. - Useful in certain physics-based data domains such as NIR images, medical imaging, and astronomical imaging. - Since CUT is a general-purpose image-to-image (I2I) model, it is widely used for visual transformations such as facial modification, landscape style transfer, and artistic rendering. - An extended CUT model with added physics-based correction loss $\mathcal{L}(G, F, D_A, D_B) = \mathcal{L}_{GAN}(G, D_B) + \lambda \mathcal{L}_{PatchNCE} + \lambda_{domain} \mathcal{L}_{NIRdomain}$ - Addition of a physics-based correction loss term : $\lambda_{domain} \mathcal{L}_{NIRdomain}$ $\mathcal{L}_{NIRdomain}$: histogram loss, smoothness loss, DCT loss, etc. λ_{domain} : control the relative importance (weights) of each loss term - Physics-based correction loss presents a promising option for improving CUT's performance, but if added without careful consideration, it may compromise the model's core strengths. - To ensure a safe experimental design, it is important to gradually increase the value of λ_{domain} during training. - Performance experiments were conducted by adjusting the value of λ_{domain} in the following order: 0.0, 0.1, 0.2, 0.5, and 0.1. - Implementation of a New Image Generation Model Based on CUT and Generation of Fake Images - Experience of multiple λ_{domain} on Colab - Set Colab Environment

- Clone CUT storage : !git clone https://github.com/taesungp/contrastive-unpaired-translation.git
%cd contrastive-unpaired-translation
- Install necessary packages : !pip install -q dominate visdom
- ii. Google Drive mount and link data root
from google.colab import drive
drive.mount('/content/drive')
csv_path = "/content/drive/MyDrive/datasets/apple2nir/Brix_nir_classified.csv"
img_dir = "/content/drive/MyDrive/datasets/apple2nir/trainB/"
- iii. Add Domain Loss function
import torch.nn as nn
import torch
class HistogramLoss(nn.Module):
def __init__(self):
super(HistogramLoss, self).__init__()
def forward(self, fake, real):
loss = 0
for i in range(fake.shape[1]):
f_hist = torch.histc(fake[:, i], bins=256, min=0.0, max=1.0)
r_hist = torch.histc(real[:, i], bins=256, min=0.0, max=1.0)
f_hist = f_hist / f_hist.sum()
r_hist = r_hist / r_hist.sum()
loss += torch.mean((f_hist - r_hist) ** 2)
return loss
- iv. Add the lambda_domainoption in train.py
parser.add_argument('--lambda_domain', type=float, default=0.0, help='weight for domain-specific loss')
if opt.lambda_domain > 0:
domain_loss = histogram_loss(fake_B, real_B) * opt.lambda_domain
loss_G += domain_loss
- v. Conduct repeated training for various λ values
lambda_list = [0.0, 0.1, 0.2, 0.5, 1.0]

for lam in lambda_list:
experiment_name = f"cut_lambda_{lam}"
print(f" Running: lambda_domain = {lam}")

!python train.py --dataroot ./datasets/apple2nir \
--name {experiment_name} \
--CUT_mode CUT \
--lambda_domain {lam} \
--n_epochs 20 \
--batch_size 4 \
--gpu_ids 0
- vi. Run test.py for each λ value
!python test.py --dataroot ./datasets/apple2nir \
--name cut_lambda_0.0 \
--CUT_mode CUT \
--phase test \
--no_dropout \
--num_test 9999 \
--epoch latest \
--gpu_ids 0

3. Performance analysis and evaluation using the EfficientNet classification model

λ_{domain}	Accuracy	Low F1	Mid F1	High F1
0.0	30%	0.00	0.00	0.50
0.1	20%	0.00	0.33	0.25
0.2	40%	0.00	0.00	0.62
0.5	70%	0.80	0.80	0.40
1.0	50%	0.67	0.62	0.00

- A. $\lambda_{domain} = 0.5$ showed the best performance. NIR characteristics such as brightness patterns and tissue structure were most clearly observed in the Low and Mid classes.
- B. Using the extended CUT model with physics-based correction loss and $\lambda_{domain} = 0.5$, we aim to convert a large set of apple RGB images (from the Jeonbuk Jangsu apple sugar contents quality dataset on AI Hub) into NIR images.
- C. We aim to develop a classification model based on these generated images.

4. Reflection and Conclusion

- i. In previous studies, CycleGAN, CUT, and MUNIT models were applied to RGB-to-NIR image translation tasks, and among them, the CUT model demonstrated the best classification performance. However, its performance was still insufficient. To improve upon this, extended CUT-based models such as DCLGAN and DualCUT were tested. Nevertheless, these models performed worse or comparable to the original CUT, failing to show significant improvements.
- ii. Accordingly, this study focused again on the CUT model, maintaining its core architecture while introducing a **Domain-Specific Loss** to better reflect the physical characteristics of NIR images.
- iii. The additional loss components introduced are as follows:
 - **Histogram Matching Loss:** Reflects the average brightness distribution of NIR images
 - **Reflectance Smoothness Loss:** Preserves the internal structural consistency of fruit
 - **DCT/Frequency Loss:** Captures high-frequency reflectance characteristics
- iv. By integrating these losses, the model was trained with varying λ values from 0.0 to 1.0. The best classification performance was achieved at $\lambda = 0.5$, recording an accuracy of 70%. Notably, F1-scores of approximately 0.8 were observed in the Low and Mid classes, indicating that internal patterns and brightness distributions carry meaningful information for sugar contents classification.

These results suggest that combining CUT's structural preservation capability with NIR-specific loss functions enables the generation of images that capture not only visual similarity but also spectrally meaningful features.

17.Date: 2025 May

Participants: Koh Youn Seo, Kim Seung Geon

Topic: Implementation of Apple sugar contents Classification Model Using an Extended CUT Model with Physics-Based Correction Loss

Objective:

- Preprocessing RGB data of the Fuji variety from *Jeonbuk Jangsu Apple sugar contents Quality Dataset*
- Generating fake NIR images using an extended CUT model with physics-based correction loss
- Performance analysis and evaluation using an EfficientNet classification model

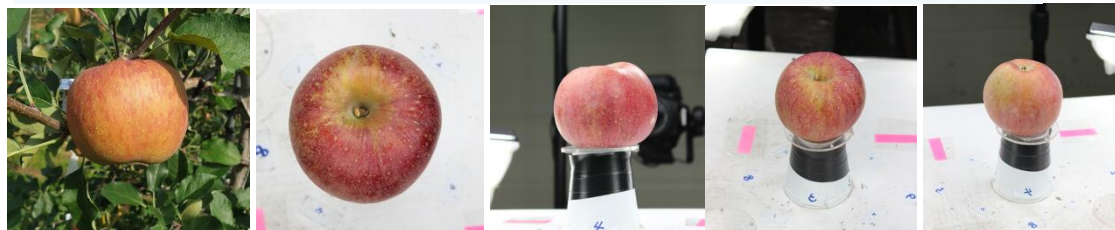
Contents:

1. Dataset Source

A. Jeonbuk Jangsu apple sugar contents quality data from <https://www.aihub.or.kr/>

1. Training data - 2D RGB image data - Construction scale

Crop name	Non-destructive grade	Data count	ratio
Shinano Gold	A	17,207	3%
	B	20,581	4%
	C	5,586	1%
	subtotal	43,374	8%
Arisoo	A	11,651	2%
	B	22,692	4%
	C	7,787	1%
	subtotal	42,130	8%
Hongro	A	66,309	12%
	B	95,138	18%
	C	29,315	5%
	subtotal	190,762	36%
Fuji	A	127,509	24%
	B	87,734	16%
	C	44,182	8%
	subtotal	259,425	48%
total		535,691	100%



B. Fuji Apple Image Data Statistics

- High sugar grade(A): 127,509 RGB images
- Medium-sugar grade(B): 87,734 RGB images

- iii. Low sugar grade(C): 44,182 RGB images
- iv. Grade definitions
 - A grade>14 Brix
 - B grade: 12-14Brix
 - C grade: <12 Brix
- C. Apple Segmentation from Background
 - i. Attempted Automatic Deletion of Top View Images - Failed
 - Top view characteristics: circular shape, stem centered, equal height and width ration, dark spot in the center.
 - Techniques used: aspect ratio filtering, Hough Circle Detection, binary classification of top/side views
 - Code was implemented, but automatic deletion of top-view images was unsuccessful
 - ii. Attempted Stem-Based Deletion of Top-View Images - Failed
 - Extracted the central 20% of the image.
 - Calculated average brightness or proportion of dark pexels in that region
 - Classified as top-view if below a certain brightness threshold
 - Color filtering based on stem characteristics was attempted, but results were not satisfactory.
 - iii. Manual Deletion of Undesired Images
 - Deleted top-view images, orchard-captured images, and those taken at 45 degree angles.



- iv. Final remaining image counts:
 - Grade A: 9079
 - Grade B: 6087
 - Grade C: 1918
- D. Cropping to Focus on Apples



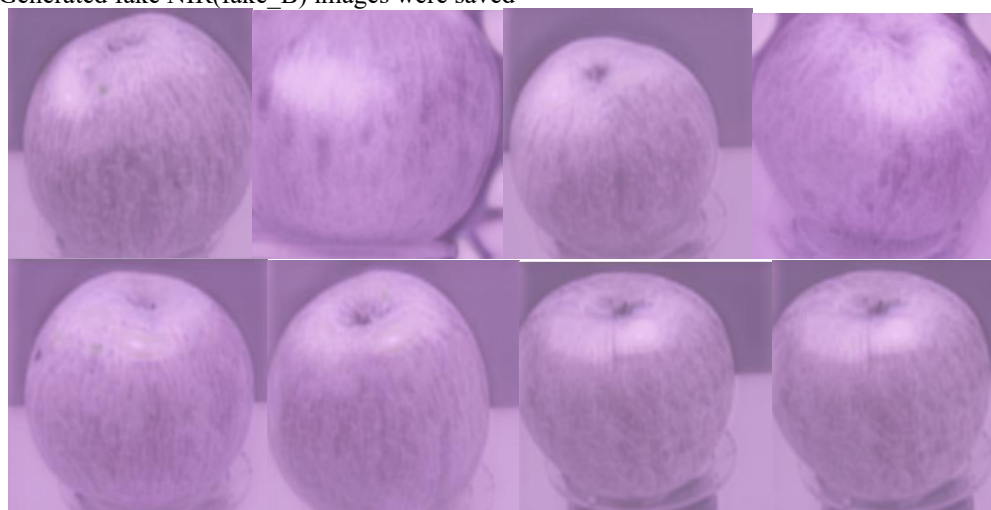
- i. Apples appeared too small due to large background areas
- ii. Applied OpenCV based center crop to remove unnecessary background
- 2. Generating Fake NIR Images Using Extended CUT Model with Physics-Based Correction Loss
 - A. Used extended CUT model with additional domain loss to generate fake NIR images from RGB images across A, B and C sugar levels
 - B. Merged A, B, C class folders into a single testA folder for testing
 - C. Created a CSV file mapping filenames to their sugar contents grades while merging folders

	A	B		A	B		A	B
1	filename	label	6515	B_20211025_RGB_mid		11636	A_20211024_RGB_high	
2	C_20211030_RGB_low		6516	B_20211025_RGB_mid		11637	A_20211024_RGB_high	
3	C_20211030_RGB_low		6517	B_20211025_RGB_mid		11638	A_20211024_RGB_high	
4	C_20211030_RGB_low		6518	B_20211025_RGB_mid		11639	A_20211024_RGB_high	
5	C_20211030_RGB_low		6519	B_20211025_RGB_mid		11640	A_20211024_RGB_high	
6	C_20211030_RGB_low		6520	B_20211025_RGB_mid		11641	A_20211024_RGB_high	
7	C_20211030_RGB_low		6521	B_20211025_RGB_mid		11642	A_20211024_RGB_high	
8	C_20211030_RGB_low		6522	B_20211025_RGB_mid		11643	A_20211024_RGB_high	
9	C_20211030_RGB_low		6523	B_20211025_RGB_mid		11644	A_20211024_RGB_high	
10	C_20211030_RGB_low		6524	B_20211025_RGB_mid		11645	A_20211024_RGB_high	
11	C_20211030_RGB_low		6525	B_20211025_RGB_mid		11646	A_20211024_RGB_high	
12	C_20211030_RGB_low		6526	B_20211025_RGB_mid		11646	A_20211024_RGB_high	

D. Model Inference Command Example

```
!python test.py \
--dataroot /content/aihub_rgb_testA \
--name cut_lambda_0.5 \
--CUT_mode CUT \
--no_dropout \
--phase test \
--epoch latest \
--num_test 999999 \
--gpu_ids 0
```

E. Generated fake NIR(fake_B) images were saved



3. Performance Analysis Using EfficientNet with 17084 Fake NIR Apple Images

A. First Classification Model Results

Class	Precision	Recall	F1-score
Low	0.85	0.53	0.66
Mid	0.78	0.91	0.84
High	0.83	0.76	0.80

- i. Accuracy : 80%
- ii. Low Class (Low Sugar)
 - High precision(0.85): 85% of predicted “Low” apples were correct
 - Low recall(0.53): Missed many actual “Low” class samples
 - Likely due to significant class imbalance(very few Low-class samples)
- iii. Mid Class(Mid Sugar)
 - High recall(0.91): Most Mid samples were correctly identified
 - Highest F1 score(0.84) among all classes
 - Performance attributed to the large amount of training data
- iv. High Class (High Sugar)
 - High precision(0.83) and recall(0.76) : Stable and reliable predictions

B. EfficientNet Model Evaluation

Model	Accuracy	Low F1	Mid F1	High F1
physCUT	80%	0.66	0.80	0.80

- i. **Accuracy of 80%** indicates that the generated images are of sufficient quality and resemble real NIR images
- ii. **Mid Class:** Well-predicted due to the abundance of training data (Recall 91%)
- iii. **High Class:** Stable classification with precision of 83% and recall of 76%
- iv. **Low Class:** Despite high precision (85%), the recall remains low (53%)—likely caused by underfitting or class imbalance due to a smaller number of samples

4. Reflection and Conclusion

- i. This study began with the principle of spectroscopy for classifying fruit sugar contents. Initially, an experiment was conducted using a dataset of 60 apples captured with both RGB and NIR images to classify sugar contents based on NIR information. However, due to the limitations of small-scale data and the high cost of NIR equipment, we explored an alternative approach that generates NIR-style images from RGB images using a large-scale public dataset (AiHub), aiming to predict sugar contents indirectly.
- ii. Based on this background, we designed an extended model of the CUT framework by integrating three Domain-Specific Loss functions-Histogram Matching, Reflectance Smoothness, and Frequency Loss-to better reflect the physical characteristics of the NIR domain. Using this model, a total of 17,084 fake NIR images were generated from the AiHub RGB apple images, and an EfficientNet classifier was trained to predict apple sugar contents levels (Low, Mid, High).
- iii. The classification accuracy reached 80%, with F1-scores of 0.84 for the Mid class, 0.80 for the High class, and 0.66 for the Low class. The Mid class achieved stable predictions due to its larger data proportion, while the Low class showed a lower recall of 0.53, likely due to class imbalance and limited data.

In conclusion, Domain-Specific Loss-based CUT extension model effectively combines the structural preservation capability of the original CUT with the physical properties of the NIR domain. This experimental approach demonstrates the potential of indirect spectral analysis without NIR sensors and provides meaningful validation of the research's value as a non-destructive quality evaluation technique.